

Worksheet

1. Does the sequence $\left\{ \left(\frac{4}{5} \right)^k \right\}$ converge? If so what is the limit? If not, why not?
2. Does the series $\sum_{k=0}^{\infty} \left(\frac{4}{5} \right)^k$ converge? If so what does it converge to? If not, why not?
3. Does the series $\sum_{k=2}^{\infty} 3 \left(\frac{4^{k+2}}{5^k} \right)$ converge? If so what does it converge to? If not, why not?
4. Does $\sum_{k=3}^{\infty} \ln \left(\frac{n}{n+1} \right)$ converge? Why or why not?
5. Does $\sum_{k=1}^{\infty} 5 \frac{3^{k-2}}{2^{k+1}}$ converge? If so what does it converge to? If not, why not?
6. Does $\sum_{k=1}^{\infty} 5 \frac{2^{k-2}}{7^{k+1}}$ converge? If so what does it converge to? If not, why not?

Do the following series converge or diverge? Be sure to state any tests used and show all work needed apply them.

7. $\sum_{k=0}^{\infty} \frac{4k}{5k-3}$
8. $\sum_{k=1}^{\infty} \frac{\ln(k^2)}{k}$
9. $\sum_{k=1}^{\infty} \sqrt[k]{k}$
10. $\sum_{k=1}^{\infty} \frac{3}{k \sqrt[3]{k}}$